3 hours on examples

Problem 1 (13INMO3)

FTSOC, let there be an integer solution k to the polynomial. We know,

$$k^4 - ak^3 - bk^2 - ck - d = 0$$

$$\Leftrightarrow k(k(k(k-a)-b)-c)-d = 0$$

By the integer root theorem, we know $k \mid d$ meaning $k \leq d \leq c \leq b \leq a$. Therefore,

$$k - a \leq 0$$

$$\Rightarrow k(k-a) - b < 0$$

$$\ldots$$

$$\Rightarrow \ k(k(k(k-a)-b)-c)-d < 0 \ (!)$$

Problem 2 (04TTSSA5)

We claim that it is possible using the parabola $y = x^2$ and circle $(x+4)^2 + \left(y - \frac{7}{2}\right)^2 = \frac{125}{4}$.

Rotate and translate the diagram such that the parabola so that its axis of symmetry is parallel with the y-axis and vertex at $(0,0)$. Note that the circle will still be a circle after the rotation. We now graph the two conics on the xy-plane and express their formulas.

$$y = px^2$$

$$r^2 = (x-m)^2 + (y-n)^2$$

Plugging y in,

$$0 = (x-m)^2 + (px^2 - n)^2 - r^2$$

There should only be two roots to this equation at the x-coordinates of A and B respectively. These two coordinates are distinct since $y = ax^2$ isa function.

We are given that the root at A is at least a double root. Since the degree is four, it is impossible for the root at A to be a quadruple root. We have our equation can be written as either one of the following, for some $a \neq b$,

$$(x-a)^3(x-b) = (x-m)^2 + (px^2 - n)^2 - r^2$$

$$(x-a)^2(x-b)^2 = (x-m)^2 + (px^2 - n)^2 - r^2$$

We want to find a counterexample belonging to the first case. We force $p = 1$,

$$(x-a)^3(x-b) = (x-m)^2 + (x^2-n)^2 - r^2$$

We want an inflection point at $x = a$ so we take the double derivative of the RHS,

$$2 + 12a^2 - 4n = 0 \Leftrightarrow a = \sqrt{\frac{2n-1}{6}}$$

We take $n = \frac{7}{2}$ such that $a = 1$.

$$(x-1)^3(x-b) = (x-m)^2 + \left(x^2 - \frac{7}{2}\right)^2 - r^2$$

Taking the first derivative at $x = 1$,

$$0 = 2(1-m) + 4\left(1 - \frac{7}{2}\right) \Leftrightarrow m = -4$$

We now plug in $x = 1$ to get $0 = 5^2 + \frac{25}{4} - r^2 \Leftrightarrow r^2 = \frac{125}{4}$ to finish. We may verify the construction by plugging everything in.

4 clubs; 4 hours

Problem 3 (13SMTA9)

Let the desired sum be s,

$$s = -\frac{\sum_{cyc} a^4(b-c)}{(a-b)(b-c)(c-a)}$$

We factor the top. Note that plugging in any pair of a, b, c equal into the top, it disappears meaning $(a-b)(b-c)(c-a)$ is a factor of $\sum_{cyc} a^4(b-c)$.

Since the top is a degree-5 symmetric sum, we write it as,

$$\sum_{cyc} a^4(b-c) = (a-b)(b-c)(c-a)\big(k_1(a^2+b^2+c^2) + k_2(ab+bc+ca)\big)$$

Plugging in $(a, b, c) = (-2, 3, 6)$, we have,

$$16(-3) + 81(8) + 1296(-5) = (-5)(-3)(8)(k_1(4+9+36) + 0)$$

$$\Leftrightarrow -6 + 81 - 810 = 15(49)k_1$$

Therefore, $k_1 = -1$. Now plugging in $(a, b, c) = (-1, 0, 1)$, we have,

$$-1 - 1 = (-1)(-1)(2)\big(k_1(2) + k_2(-1)\big)$$

Therefore, $k_2 = -1$ as well. In conclusion,

$$\sum_{cyc} a^4(b-c) = -(a-b)(b-c)(c-a)(a^2+b^2+c^2+ab+bc+ca)$$

$$\Leftrightarrow s = -\frac{\sum_{cyc} a^4(b-c)}{(a-b)(b-c)(c-a)} = a^2+b^2+c^2+ab+bc+ca$$

$$= \frac{1}{2}((a+b)^2 + (b+c)^2 + (c+a)^2)$$

$$= \frac{1}{2}\left(\left(2\sqrt{7}\right)^2 + \left(2\sqrt{3}\right)^2 + \left(2\sqrt{5}\right)^2\right) = 14 + 6 + 10 = 30$$

<u>Problem 4 (11SMTA7)</u>

Note that, for $x = 1, 2, \dots, 2^{2010}$,

$$P(2x) - P(x) = 1$$

We let $Q(x) = P(2x) - P(x) - 1 = a \cdot (x-1)(x-2)\dots(x-2^{2010})$. Plugging in $x = 0$, we have,

$$Q(0) = -1 = a \cdot (-1)(-2)\dots(-2^{2010})$$

$$\Leftrightarrow a = \frac{1}{2^{0+1+\dots+2010}}$$

Note that the coefficient of x in P is the same of that in Q.

$$Q(x) = \frac{1}{2^{0+1+\dots+2010}}(x-1)(x-2)\dots(x-2^{2010})$$

The desired coefficient would be,

$$1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^{2010}} = 2 - \frac{1}{2^{2010}}$$

9 clubs; 5 hours

Problem 6 (23INMO2)

FTSOC, let this be possible.

Claim: All roots of all polynomials of the form $a_k x^{100} + \cdots + a_{98+k}x^2 + 100a_{99+k}x + a_{100+k}$ are real roots.

Proof: Since $a_k \neq 0$, we know 0 was not a root to the previous equation. By reversing the coefficients, the roots are now the reciprocals of what they were before. □

Lemma: If $P(x)$ has all real roots, so does $P'(x)$.

Proof: Let P have roots $r_1, \dots, r_k$ with multiplicities $m_1, \dots, m_k$. We know that $P'(x)$ still has roots $r_1, \dots, r_k$ with multiplicities $m_1 - 1, \dots, m_k - 1$ for those $m_i > 1$. Furthermore, by the Mean Value Theorem, there is another instance where $P'(x) = 0$ between every root. □

We take the 98-th derivative of $a_k x^{100} + \cdots + a_{98+k}x^2 + 100a_{99+k}x + a_{100+k}$ to get a quadratic with real roots.

$$\frac{100!}{2} a_k x^2 + 99!\, a_{k+1} x + 98!\, a_{k+2}$$

Looking at the discriminant,

$$(99!\, a_{k+1})^2 - 4 \cdot \frac{100!}{2} \cdot 98!\, a_k a_{k+2}$$

$$\Leftrightarrow 99 a_{k+1}^2 \geq 200\, a_k a_{k+2}$$

This is clearly impossible if we take $a_{k+1} = \min(a_0, \dots, a_{100})$.

Problem 7 (56PTNMB7)

We write,

$$(x - r_1)^{m_1}(x - r_2)^{m_2} \dots (x - r_k)^{m_k} = P(z)$$

$$(x - r_1)^{n_1}(x - r_2)^{n_2} \dots (x - r_k)^{n_k} = Q(z)$$

$$(x - r_1')^{m_1'} \dots (x - r_l')^{m_l'} = P(z) + 1$$

$$(x - r_1')^{n_1'} \dots (x - r_l')^{n_l'} = Q(z) + 1$$

Note that $P(z) - Q(z)$ has roots at $r_1, \dots, r_k$ and $r_1', \dots, r_l'$. We cannot have $r_i = r_j'$ since then $P(r_i) = 0$ and $P(r_j') = -1$. Therefore, $\deg(P(z) - Q(z)) \geq l + k$. WLOG, let $\deg(P) \geq \deg(Q)$. We know that $\deg(P) \geq l + k$.

We now differentiate $P(x)$. We know,

$$P'(x) = \frac{d}{dx}((x - r_1)^{m_1} \dots (x - r_k)^{m_k}) = \frac{d}{dx}\left((x - r_1')^{m_1'} \dots (x - r_l')^{m_l'}\right)$$

Of the roots $r_1, \dots, r_k$, all of them who had multiplicity > 1 are still roots of P with multiplicity one less than before. Therefore, they contribute at least $\deg(P) - k$ to the degree of P'.

The same applies to the roots $r_1', \dots, r_l'$. Since these roots are distinct from $r_1, \dots, r_k$, they contribute another $\deg(P) - l$ to the degree of P'.

$$\deg(P') = \deg(P) - 1 \geq 2\deg(P) - k - l$$

$$\Leftrightarrow \deg(P) \leq k + l - 1$$

This contradicts our previous finding, so we are done.

14 clubs; 8 hours

Problem 8 (Z6597084)

We claim that $P(x) = (x^2 + 1)^k$ for some integer $k \geq 0$. Plug in to check these work.

If P is constant, clearly $P = 1$. Otherwise, P has a root and leading coefficient still 1.

We see that, if x is root to our polynomial, plugging in x and $x - 1$ gives $x^2 \pm x + 1$ are both roots as well. Let X be the root to our polynomial of maximal magnitude. Since $X^2 \pm X + 1$ are both roots as well,

$$|X| \geq |X^2 + X + 1|, |X^2 - X + 1|$$

$$\Longrightarrow 2|X| \geq |X^2 + X + 1| + |X^2 - X + 1|$$

The equality is true if and only if $|X| = |X^2 + X + 1| = |X^2 - X + 1|$.

However, by the triangle inequality,

$$|X^2 + X + 1| + |X^2 - X + 1| \geq |2X| = 2|X|$$

The equality case is true only if $X^2 + X + 1, X^2 - X + 1$ have opposite directions.

Since the two inequalities have different directions, the equality case is true in both inequalities.

From $X^2 + X + 1, X^2 - X + 1$ in opposite directions, $X^2 + X + 1 = -X^2 + X - 1$ meaning $X^2 = -1$. The maximum magnitude root is then $\pm i$.

As a result, no root has magnitude > 1, and the only roots with magnitude 1 are exactly $\pm i$.

Plugging in $x = 1$, we see that $P(1)\,P(0) = P(1)$. Since $P(1) \neq 0$, $P(0) = 1$ meaning the product of the roots is 1. Therefore, since there cannot be roots with magnitude greater than 1, all the roots have magnitude 1. As a result, all the roots must be $\pm i$. These must in conjugate pairs as desired.

<u>Problem 9 (22PTMA2)</u>

Of the $2n + 1$ coefficients of $p(x)^2$, we claim that at most $2n - 2$ of them are negative.

Note that scaling $p(x)$ by r changes each term of $p(x)^2$ by $r^2 > 0$. Therefore, WLOG scale $p(x)$ such that its leading coefficient is $a_n := 1$.

$$p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$$

Lemma: If the coefficients for $x^{2n-1}, \dots, x^n$ are negative for $p(x)^2$, all the non-leading coefficients of $p(x)$ are negative. It is also possible for the coefficients of $x^{2n-1}, \dots, x^n$ of $p(x)^2$ to be any negative integer for some $p(x)$.

We use strong induction to prove that $a_i < 0$ for all i.

Base case: $i = n - 1$. The coefficient of x^{2n-1} is negative in $p(x)^2$ so a_{n-1} must be negative.

Inductive step: $i \to i - 1$. The coefficient of $n + i - 1$ is,

$$a_n a_{i-1} + (a_{n-1}a_i + \cdots + a_i a_{n-1}) + a_{i-1}a_n$$

Each term of $(a_{n-1}a_i + \cdots + a_i a_{n-1})$ is positive since it is the product of two negative numbers. For the overall sum to be negative, $a_{i-1}a_n = a_n a_{i-1}$ must be negative as a result. Since $a_n = 1$ is positive, $a_n < 0$ as desired.

Furthermore, it is also possible to construct a_{i-1} such that the quantity is any arbitrary negative integer since $a_n \neq 0$ and $(a_{n-1}a_i + \cdots + a_i a_{n-1})$ is finite. □

We first prove the bound. Clearly, the leading coefficient and constant term of $p(x)^2$ cannot be negative. FTSOC, let all the remaining coefficients of $p(x)^2$ be negative. By the lemma, each non-leading coefficient of $p(x)$ is negative. In this case, however, the coefficient of x in $p(x)$ is positive. (!)

We now show the construction for $2n - 2$ negatives. Using the lemma, we may first construct a $q(x)$ such that $q(x)^2 = x^{2n} - x^{2n-1} - \cdots - x^n + a_{n-1}x^{n-1} + \cdots + a_0$ for some integers $a_{n-1}, \dots, a_0$. By the lemma, $q(x) = x^n + q_{n-1}x^{n-1} + \cdots + q_0$ for negative coefficients $q_0, \dots, q_{n-1}$.

We now plug in, $p(x) = q(x) + c$ for some integer c.

$$p(x)^2 = q(x)^2 + 2c\,q(x) + c^2$$

$$= x^{2n} - x^{2n-1} - \cdots - x^n + a_{n-1}x^{n-1} + \cdots + a_0$$

$$+2cx^n + 2cq_{n-1}x^{n-1} + \cdots + 2cq_0$$

$$+c^2$$

Since each of $q_{n-1}, \dots, q_1$ is negative, for a sufficiently large c, we may ensure the coefficients of $x^{n-1}, \dots, x$ in $p(x)$ is negative. Furthermore, the coefficients of $x^{2n-1}, \dots, x^{n+1}$ are still unaffected at -1. In total, we have $2n - 2$ negative coefficients as desired.

22 clubs; 10 hours

<u>Problem 12 (24VNM5)</u>

We claim the answer is yes. The following polynomial works,

$$P(x) = x^2 - a^2$$

We induct to see that $P^{(n)}(x) = t$ has distinct real solutions $x = \pm\sqrt{a^2 \pm \sqrt{\dots\sqrt{a^2 \pm \sqrt{a^2 + t}}}}$ with n iterations satisfying $x \in (-a-1, a+1)$.

Base case: $n = 1$. We have $x^2 - a^2 = t \Leftrightarrow x = \pm\sqrt{a^2 + t}$. Since $-a < t < a$, we have the inequality $|x| < \sqrt{a^2 + a} < a + 1$ as desired. We also see that $a^2 + t > 0$ from $a \geq 2$ and $-a < t < a$ so this is real.

Inductive step: Given that $P^{(k)}(x)$ has solutions $x = \pm\sqrt{a^2 \pm \sqrt{\dots\sqrt{a^2 \pm \sqrt{a^2 + t}}}}$ we k iterations, we see that,

$$P^{(k)}\big(P(x)\big) = t \Leftrightarrow P(x) = \pm\left(\sqrt{a^2 \pm \sqrt{\dots\sqrt{a^2 + t}}}\right)$$

$$\Leftrightarrow x = \pm\sqrt{a^2 \pm \left(\sqrt{a^2 \pm \sqrt{\dots\sqrt{a^2 + t}}}\right)}$$

We now check that, since $\pm\left(\sqrt{a^2 \pm \sqrt{\dots\sqrt{a^2+t}}}\right) \in (-a-1, a+1)$, we have that,

$$a^2 - a - 1 < a^2 \pm \left(\sqrt{a^2 \pm \sqrt{\dots\sqrt{a^2+t}}}\right) < a^2 + a + 1$$

Indeed, the lowerbound is positive and the upper-bound is less than $(a+1)^2$.

Finally, note that if the previous solutions are distinct, $\sqrt{a^2 \pm \left(\sqrt{a^2 \pm \sqrt{\dots\sqrt{a^2+t}}}\right)}$ are all distinct positive real numbers meaning adding a $\pm$ and they will still be distinct.

Plugging in $n = 2024$, there are 2^{2024} solutions since 2024 $\pm$s.

<u>Problem 15 (19SLA5)</u>

We multiply both sides by $\left(\prod_{1\le k<l\le n} x_k - x_l\right) \neq 0$ to see that it suffices to prove,

$$\sum_{1\le i\le n} (-1)^{i-1}\left(\prod_{\substack{j\neq i\\ 1\le j\le n}} 1 - x_i x_j\right)\left(\prod_{\substack{1\le k<l\le n\\ k,l\neq i}} x_k - x_l\right) = \begin{cases} 0 \text{ if } n \text{ is even} \\ \left(\prod_{1\le k<l\le n} x_k - x_l\right) \text{ if } n \text{ is odd}\end{cases}$$

We show that the following multi-variable polynomials for n even and odd respectively both simply to zero,

- $\sum_{1\le i\le n}(-1)^{i-1}\left(\prod_{\substack{j\neq i\\ 1\le j\le n}} 1 - x_i x_j\right)\left(\prod_{\substack{1\le k<l\le n\\ k,l\neq i}} x_k - x_l\right)$ for even n
- $\sum_{1\le i\le n}(-1)^{i-1}\left(\prod_{\substack{j\neq i\\ 1\le j\le n}} 1 - x_i x_j\right)\left(\prod_{\substack{1\le k<l\le n\\ k,l\neq i}} x_k - x_l\right) - \left(\prod_{1\le k<l\le n} x_k - x_l\right)$ for odd n

We induct on n to prove this.

Base case: $n = 1$. Trivially true. We note that $\prod$ of no terms equals one.

Inductive step: Call the polynomial P. We have $\deg(P) = 2(n-1) + \binom{n}{2} - (n-1) = \frac{n^2+n-2}{2}$ in both cases. However, we show that,

- $x_i - 1$ is a factor of the polynomial for all i
- $x_i - x_j$ is a factor of the polynomial for all $n \ge i > j \ge 1$

This would imply that $(\prod_{1\le k<l\le n} x_k - x_l)(\prod x_i)$, a degree $\frac{n^2+n}{2}$ polynomial is a multiple of P. Hence, this would suffice to show $P \equiv 0$.

To start by showing $x_a - x_b$ is a factor, we plug in $x_a = x_b$ for any $a \neq b$. In the odd case, $(\prod_{1\le k<l\le n} x_k - x_l) = 0$ so in either cases we may only consider,

$$\sum_{1\le i\le n} (-1)^{i-1} \left(\prod_{\substack{j\neq i \\ 1\le j\le n}} 1 - x_i x_j \right) \left(\prod_{\substack{1\le k<l\le n \\ k,l\neq i}} x_k - x_l \right)$$

For all $i \neq a, b$, we have $x_a = x_b$ causes $\left(\prod_{\substack{1\le k<l\le n \\ k,l\neq i}} x_k - x_l \right) = 0$. Therefore, we show that, when $x_a = x_b$,

$$(-1)^{a-1} \left(\prod_{\substack{j\neq a \\ 1\le j\le n}} 1 - x_a x_j \right) \left(\prod_{\substack{1\le k<l\le n \\ k,l\neq a}} x_k - x_l \right) + (-1)^{b-1} \left(\prod_{\substack{j\neq b \\ 1\le j\le n}} 1 - x_b x_j \right) \left(\prod_{\substack{1\le k<l\le n \\ k,l\neq b}} x_k - x_l \right) = 0$$

$$\Leftrightarrow \frac{\left(\prod_{\substack{j\neq a \\ 1\le j\le n}} 1 - x_a x_j \right) \left(\prod_{\substack{1\le k<l\le n \\ k,l\neq a}} x_k - x_l \right)}{\left(\prod_{\substack{j\neq b \\ 1\le j\le n}} 1 - x_b x_j \right) \left(\prod_{\substack{1\le k<l\le n \\ k,l\neq b}} x_k - x_l \right)} = (-1)^{a-b-1}$$

Since $x_a = x_b$, $\frac{\left(\prod_{\substack{j\neq a \\ 1\le j\le n}} 1 - x_a x_j \right)}{\left(\prod_{\substack{j\neq b \\ 1\le j\le n}} 1 - x_b x_j \right)} = 1.$

$$\Leftrightarrow \frac{\prod_{\substack{1\le k<l\le n \\ k,l\neq a}} x_k - x_l}{\prod_{\substack{1\le k<l\le n \\ k,l\neq b}} x_k - x_l} = (-1)^{a-b-1}$$

We factor $\prod_{\substack{1\le k<l\le n \\ k,l\notin\{a,b\}}} x_k - x_l$ out from top and bottom. WLOG, let $b > a$,

$$\Leftrightarrow \frac{(x_1 - x_b) \dots (x_{a-1} - x_b)(x_{a+1} - x_b) \dots (x_{b-1} - x_b)(x_b - x_{b+1}) \dots (x_b - x_n)}{(x_1 - x_a) \dots (x_{a-1} - x_a)(x_a - x_{a+1}) \dots (x_a - x_{b-1})(x_a - x_{b+1}) \dots (x_a - x_n)} = (-1)^{a-b-1}$$

Since $x_a = x_b$, the only terms different are $\frac{x_{a+1}-x_b}{x_a - x_{a+1}} \cdot \dots \cdot \frac{x_{b-1}-x_b}{x_a - x_{b-1}} = \frac{x_{a+1}-x_a}{x_a - x_{a+1}} \cdot \dots \cdot \frac{x_{b-1}-x_a}{x_a - x_{b-1}}$. Each of these are -1 so their product is $(-1)^{a-b-1}$ as desired.

We now show that each $x_a - 1$ is a factor by plugging in $x_a = 1$. We may now divide back out what we initially multiplied on. We show that, when $x_a = 1$,

$$\sum_{1\le i\le n} \prod_{\substack{j\neq i \\ 1\le j\le n}} \frac{1 - x_i x_j}{x_i - x_j} = \begin{cases} 0 \text{ if } n \text{ is even} \\ 1 \text{ if } n \text{ is odd} \end{cases}$$

$$\Leftrightarrow \left(\sum_{\substack{1 \le i \le n \\ i \ne a}} \prod_{\substack{j \ne i \\ 1 \le j \le n}} \frac{1 - x_i x_j}{x_i - x_j} \right) + \prod_{\substack{j \ne a \\ 1 \le j \le n}} \frac{1 - x_a x_j}{x_a - x_j} = \begin{cases} 0 \text{ if } n \text{ is even} \\ 1 \text{ if } n \text{ is odd} \end{cases}$$

$$\Leftrightarrow \left(\sum_{\substack{1 \le i \le n \\ i \ne a}} \left(\prod_{\substack{j \ne i,a \\ 1 \le j \le n}} \frac{1 - x_i x_j}{x_i - x_j} \right) \left(\frac{1 - x_i x_a}{x_i - x_a} \right) \right) + 1 = \begin{cases} 0 \text{ if } n \text{ is even} \\ 1 \text{ if } n \text{ is odd} \end{cases}$$

By the inductive hypothesis, $\left(\sum_{\substack{1 \le i \le n \\ i \ne a}} \left(\prod_{\substack{j \ne i,a \\ 1 \le j \le n}} \frac{1 - x_i x_j}{x_i - x_j} \right) \left(\frac{1 - x_i x_a}{x_i - x_a} \right) \right) = \begin{cases} 0 \text{ if } n \text{ is even} \\ -1 \text{ if } n \text{ is odd} \end{cases}$ so we are done.

30 clubs; 12.5 hours

Problem 16 (02AMO3)

We prove the stronger conclusion that this is true for non-monic polynomials as well. This implies the desired result since we can just scale the two polynomials we constructed accordingly.

Our construction is as follows. Let M be the maximum of $|P(x)|$ on $[0, n+1]$.

Consider the polynomial.

$$A(x) = (x-1)(x-2)\ldots(x-n)$$

We plot A on the xy-plane. By differentiation, A has $n-1$ local minimums or maximums. Hence, there is exactly one local extrema on each of $(1,2), \ldots, (n-1, n)$ since one exists at each interval by the Mean-Value theorem. The signs of the values of A at these local extrema alternate between positive and negative since A crosses the x-axis at exactly $x = 1, \ldots, n$.

Of the extrema, consider the absolute values of the y to be $y_1, y_2, \ldots, y_{n-1}$. Let,

$$M_A = \min(|y_1|, |y_2|, \ldots, |y_{n-1}|, |A(0)|, |A(n+1)|)$$

We take the two polynomials,

$$P_1(x) = -c\, A(x) + P(x)$$

$$P_2(x) = c\, A(x) + P(x)$$

For sufficiently large c such that $cM_A > M$, we see that both P_1, P_2's signs at $x = 0, n+1$ and the x-coordinates of the original extremas do not change meaning both have n real zeroes.

Problem 18 (16TTFSA6)

We claim the minimum amount is four rubles.

To prove that Vasya can win with four rubles, we may prove the following two claims:

1. Whenever Petya answers 3 or more, Vasya may force Petya to answer zero on the next two guesses.

   Proof: Given that Vasya guessed a to get an answer of 3 or more, we have,

   $$P(x) = (x - r_1)(x - r_2)(x - r_3)Q(x) + a$$

   Vasya may next guess $a \pm 1$. For $P(x) = a \pm 1$, we must have,

   $$|(x - r_1)(x - r_2)(x - r_3)Q(x)| = 1$$

   Since each factor is an integer,

   $$|x - r_1| = |x - r_2| = |x - r_3| = |Q(x)| = 1$$

   It is impossible for x to be one apart from each of the distinct integers r_1, r_2, r_3. □

2. Whenever Petya answers 2, Vasya may force Petya to answer 1 and 0 in some order over the next two guesses.

   Proof: Given that Vasya guessed a to get an answer of 2, we have,

   $$P(x) = (x - r_1)(x - r_2)Q(x) + a$$

   Vasya may next guess $a \pm 1$. For $P(x) = a \pm 1$, we must have,

   $$|(x - r_1)(x - r_2)Q(x)| = 1$$

   Since each factor is an integer, this is true if and only if,

   $$|x - r_1| = |x - r_2| = |Q(x)| = 1$$

   The only way this can be possible is if $|r_1 - r_2| = 2$ and $x = \frac{r_1+r_2}{2}$. Therefore, the only possible root of $P(x) = a \pm 1$ is $x = \frac{r_1+r_2}{2}$. Plugging in $x = \frac{r_1+r_2}{2}$ may only yield one unique value so $P(x) = a \pm 1$ has one or zero integer solutions in some value. □

With the claims proven, we see that, if Petya answers 3 or more in the first two guesses, Vasya may make two consecutive zero guesses. Otherwise, if Petya doesn't do so in the first two guesses,

1. Petya can either answer 0 or 1 in some order. On the next move, to not repeat, Petya must answer 2 or more. Using both claims, we always have a way of guaranteeing Petya answers either 0 or 1 with the next guess.
2. Petya can either answer 0 or 2 in some order. On the next move two moves, using the second claim, we may get $0, 1$ in some order and 0 must repeat.
3. Petya can either answer 1 or 2 in some order. On the next move two moves, using the second claim, we may get $0, 1$ in some order and 1 must repeat.

To show that Vasya cannot win with three rubles, let Vasya's guesses be a, b, c in that order. With the polynomial $P(x) = (x-1)(x+1)(x^k - b + a) + a$ for sufficiently large k such that $|P(x)| > \max(|a|, |b|, |c|)$ when $|x| \geq 2$.

$P(-1) = P(1) = a; P(0) = b. P(\text{everything else})$ cannot be a, b, c. Therefore, Petya will answer $2, 1, 0$ respectively.

38 clubs; 15 hours

Problem 19 (23TSTST5)

We claim $p = -3; q = 3$ is the only solution.

Plug in $a = \frac{x}{y}; b = \frac{y}{z}; c = \frac{z}{x}$ from $abc = 1$ for some complex numbers x, y, z. The condition on a, b, c is satisfied if and only if no pair of x, y, z have the same argument or magnitude. Furthermore, clearly, none of x, y, z may have argument 0.

WLOG, scale each of x, y, z such that $xyz = 1$.

$$p = \sum_{sym} \frac{x}{y} = \frac{(x+y+z)(xy+yz+zx)}{xyz} - 3 = (x+y+z)(xy+yz+zx) - 3$$

On the other hand,

$$q = \sum_{cyc} \frac{xz}{y^2} = \frac{x^3y^3 + y^3z^3 + z^3x^3}{x^2y^2z^2}$$

$$= \frac{(xy+yz+zx)^3 - 3xyz(x+y+z)(xy+yz+zx)}{x^2y^2z^2} + 3$$

$$= (xy+yz+zx)^3 - 3(x+y+z)(xy+yz+zx) + 3$$

$$= (xy+yz+zx)^3 - 3p - 6$$

Given that both p, q are real, $(xy + yz + zx)^3 - 3p - 6$ is real if and only if $(xy + yz + zx)^3$ is real. Multiply each of x, y, z by $e^{\frac{2\pi}{3}}$ or $e^{\frac{4\pi i}{3}}$ as necessary such that $xy + yz + zx$ is real. Note that we can do this since $xyz = 1$ is unchanged.

With the updated x, y, z, we see that $p = (x + y + z)(xy + yz + zx) - 3$ is real if and only if $x + y + z$ **is also real or $\boldsymbol{xy + yz + zx = 0}$.**

If $\boldsymbol{xy + yz + zx = 0}$, then $\boldsymbol{p = -3}$ and $\boldsymbol{q = 3}$.

Otherwise, $x + y + z$ is real and by Vieta's, there must exist real numbers x, y, z with distinct magnitude and argument such that they are the roots of a real polynomial,

$$P(X) = X^3 + aX^2 + bX - 1$$

However, with real polynomials, complex roots come in conjugate pairs. Therefore, any complex root means there are two complex roots with the same magnitude which is not allowed. If all three roots are real, there are two roots with he same argument which is also not allowed.

<u>Problem 21 (19EMO2)</u>

We consider the solution is all $\theta(p) = p(\pm x - b)N(x)$ where $N(x)$ is any polynomial with no roots.

Let $\theta_0, \theta_1, \ldots$ be polynomials defined as $\theta_i = \theta(x^i)$. Using $\theta(p + q) = \theta(p) + \theta(q)$,

$$\theta(a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0) = a_n\theta_n + a_{n-1}\theta_{n-1} + \cdots + a_0\theta_0$$

Therefore, we may define the function θ based off $\theta_0, \theta_1, \ldots$ alone.

We are given that $a_n\theta_n + a_{n-1}\theta_{n-1} + \cdots + a_0\theta_0$ has an integer root if and only if $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ does.

Note that constant $f(x) = 1$ does not have integer roots so θ_0 has no integer roots.

Since $f(x) = x^k$ has an integer root for $k \geq 1$, each of $\theta_1, \theta_2, \ldots$ have an integer root.

Lemma: $\theta_n = \theta_0 P(x)$ for some integer polynomial $P(x)$ for all integers n.

Proof: From the fact that $f(x) = x^n - k^n$ always has an integer root at $x = k$, the polynomial $\theta_n - k^n\theta_0$ has a root for all integers k.

Since clearly, the same x cannot satisfy $\theta_n(x) + k^n\theta_0(x) = 0$ for two different k since $\theta_0(x) \neq 0$ on integers, we have infinitely many integers x such that $\theta_n(x) + k^n\theta_0(x) = 0$ for some k. In other words, there are infinitely many integers x such that,

$$\theta_0(x) \mid \theta_n(x)$$

We now write $\theta_n = \theta_0\, q(x) + r(x)$ after polynomial division with $\deg(r) < \deg(\theta_0)$. Therefore, for infinitely many x,

$$\theta_0(x) \mid r(x)$$

Knowing that $|r(x)| < |\theta_0(x)|$ for sufficiently large x from $\deg(r) < \deg(\theta_0)$, we see that $r(x) = 0$ for infinitely many r meaning $r(x) = 0$ as desired. □

Note that a set $\theta_0, \theta_1, \ldots$ works if and only if $\theta_0 P(x), \theta_1 P(x), \ldots$ works for any polynomial $P(x)$ with no integer roots since the $\theta(p+q) = \theta(p) + \theta(q)$ condition is homogenous and the second condition remains unchanged when we multiply on $P(x)$.

Divide $\theta_0, \theta_1, \ldots$ by θ_0 to effectively set $\theta_0 = 1$ WLOG.

Claim 1: $\theta_n = (\pm x - c)^n$ for some constant c for all positive n.

Consider the polynomial $x^n - c$ for $c \geq 1$. Clearly, this has an integer root if and only if c is an n-th power. Therefore, for positive c, $\theta_n(x) - c\,\theta_0(x) = \theta_n(x) - c$ has an integer root if and only if c is an n-th power.

Equivalently, the positive integers in the range of $\theta_n(x)$ when x is an integer is precisely the n-th powers.

Since the set of n-th powers is unbounded and $\theta_n(x)$ is a polynomial, we see that one of $\lim_{x\to\infty} \theta_n(x)$ or $\lim_{x\to-\infty} \theta_n(x)$ is $+\infty$.

If only one of $\lim_{x\to\infty} \theta_n(x)$, $\lim_{x\to-\infty} \theta_n(x)$ is positive infinity, consider the case where $\lim_{x\to\infty} \theta_n(x) = \infty$. Since θ_n is a non-constant polynomial, there is a threshold a when $\frac{d\theta_n}{dx} \geq 1$ for all $x \geq a$. $\theta_n(x)$ is also bounded above on $(-\infty, a)$. Let this bound be B. There is an integer threshold b such that $\theta_n(x) > B$ for all $x \geq b$. Consider $f(b), f(b+1), f(b+2), \ldots$ Each of these must be perfect n-th powers greater than B. Given that each perfect n-th power greater than B is in the range of f over the integers and f is strictly increasing past $x \geq a$, we see that $f(b), f(b+1), f(b+2), \ldots$ are consecutive perfect n-th powers $k^n, (k+1)^n, (k+2)^n, \ldots$ Therefore, $f(x) = (x - b + k)^n$ in this case.

If $\lim_{x\to-\infty} \theta_n(x) = \infty$, we do the analogous work to see that $f(x) = (-x + b - k)^n$ in this case.

If $\lim_{x\to\infty} \theta_n(x) = \lim_{x\to-\infty} \theta_n(x) = \infty$, we can still get degree n and leading coefficient on $[1, \sqrt{2}]$ with the same steps. Since leading coefficient is an integer, it must be 1. After which, we may repeat to get the same conclusion. □

Claim 2: Each θ_n has the same root as each other.

Proof: For some $m > n$ for m, n positive integers, FTSOC let θ_m, θ_n have distinct roots. Note that the polynomial $\theta_m - k \cdot \theta_n$ should always have a root since $x^m - k \cdot x^n = 0$ has a root at 0. However, once again let,

$$\theta_m(x) = \theta_n(x)q(x) + r(x)$$

$\theta_n = 0$ at its root only which is distinct from θ_m's root. Therefore, the roots x that make $\theta_m - k \cdot \theta_n = 0$ across all positive integers k are distinct. We repeat the steps from the lemma to suffice. Therefore, $\theta_n \mid \theta_m$. (!) □

We may finally check that the signs all match up. WLOG, let $\theta_1 = x$. FTSOC, if there exists some $\theta_n = -\theta_1^n$ for an odd n, consider the polynomial $x^n - 2x + 1$, which has a root at $x = 1$. We see that $-x^n - 2x + 1$ does not so no mismatched signs as desired.

52 clubs; 19 hours

<u>Problem 22 (11USATST3)</u>

For any $e > 0$, let $r_{x,m}$ be the remainder that $\sum_{k=0}^{m}(-1)^k \binom{m}{k} x_k$ leaves when divided by p^e. Knowing that $r_{x,m} = 0$ for all $m \geq B_x$, we consider the polynomial, which has degree $B_x - 1$.

$$P_x(n) = \sum_{k\geq 0} r_{x,m} \binom{n}{k}$$

By Newton's forward difference equation, $P_x(n) \equiv x_n \bmod p^e$ for all integers n.

Similarly, define P_y and $r_{y,m}$.

Now consider the polynomial $P(n) = P_x(n)P_y(n)$. Since polynomials $\binom{n}{0}, \binom{n}{1}, \binom{n}{2}$, … are linearly independent with increasing degree, we may find a sequence r_m such that,

$$P(n) = \sum_{k\geq 0} r_m \binom{n}{k}$$

Note that $r_m = 0$ for all $m > (B_x - 1)(B_y - 1)$ since that's the degree of $P_x(n)P_y(n)$. By Newton's forward difference equation,

$$r_m = \sum_{k=0}^{m} (-1)^k \binom{m}{k} P_x(n) P_y(n) \equiv \sum_{k=0}^{m} (-1)^k \binom{m}{k} x_n y_n \mod p^e$$

Since r_m is eventually all zero, we are done as desired.

Problem 26 (17USATST3)

FTSOC, let there be a pair of polynomials P, Q with $\gcd(P, Q) = 1$ such that,

$$P + \lambda_1 Q = R_1^2$$

$$P + \lambda_2 Q = R_2^2$$

$$P + \lambda_3 Q = R_3^2$$

$$P + \lambda_4 Q = R_4^2$$

WLOG let $\lambda_1 < \lambda_2 < \lambda_3 < \lambda_4$. Consider the equation,

$$\begin{pmatrix} 1 & 1 \\ \lambda_1 & \lambda_2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ \lambda_3 \end{pmatrix} \Leftrightarrow a = \frac{\lambda_2 - \lambda_3}{\lambda_2 - \lambda_1}; b = \frac{\lambda_3 - \lambda_1}{\lambda_2 - \lambda_1}$$

This pair a, b satisfies,

$$R_3^2 = aR_1^2 + bR_2^2$$

Note that a is negative and b is positive so we may apply difference of squares to find positive real numbers x, y such that,

$$R_3^2 = (xR_1 - yR_2)(xR_1 + yR_2)$$

Similarly, we may find positive x', y' such that $R_4^2 = (x'R_1 - y'R_2)(x'R_1 + y'R_2)$

Note that $\gcd(xR_1 - yR_2, xR_1 + yR_2) = \gcd(R_1, R_2) = 1$. Therefore, for some relatively prime polynomials P_1, P_2 where $P_1 P_2 = R_3$.

$$xR_1 - yR_2 = P_1^2;\ xR_1 + yR_2 = P_2^2 \Leftrightarrow R_1 - \frac{y}{x} R_2 = \left(\frac{P_1}{\sqrt{x}}\right)^2;\ R_1 + \frac{y}{x} R_2 = \left(\frac{P_2}{\sqrt{x}}\right)^2$$

Analogously,

$$R_1 - \frac{y'}{x'} R_2 = \left(\frac{P_3}{\sqrt{x'}}\right)^2;\ R_1 - \frac{y'}{x'} R_2 = \left(\frac{P_4}{\sqrt{x'}}\right)^2$$

We have found new pair R_1, R_2 satisfying condition... Check a few things to finish.

60 clubs; 22 hours